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easyspec News

easyspec 1.3.0.5: now also for imaging photometry

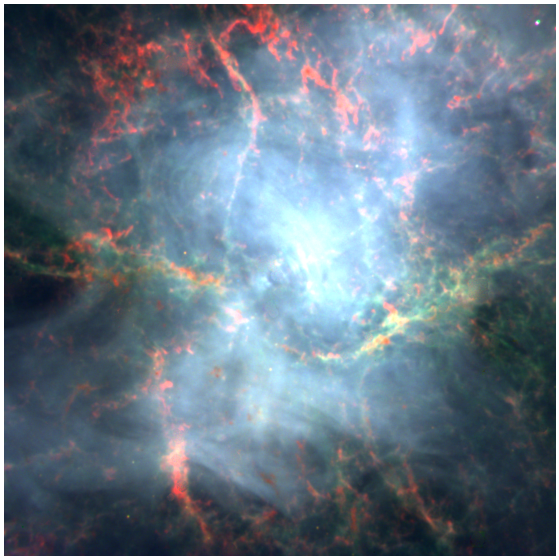
The new version of easyspec improves the extraction of spectra and goes beyond long-slit spectroscopy. The new module photometry takes you from the raw images of a field to calibrated magnitudes, light curves and surface-brightness profiles, with the same simple and explicit style as the spectroscopy modules. And with the new module gemini, raw data from the **Gemini GMOS** spectrographs can now be reduced with easyspec, in imaging and long-slit spectroscopy.

What is new in spectroscopy?

- **Optimal extraction** (Horne 1986): each pixel is weighted by the spatial profile of the source and its noise, and cosmic rays and bad pixels are rejected: a higher S/N than the simple sum, especially for faint targets.
- **Automatic identification of the arc lines**: built-in line lists of the most common lamps (HgI, NeI, ArI, HeI, CuI, CdI, ZnI, KrI, XeI, FeI and ThAr, from PypeIt) for the wavelength calibration.
- **Telluric correction**: removal of the O_2 (B and A bands) and H_2O absorption bands of the Earth's atmosphere, with the transmission measured on the standard star or taken from an external model, scaled to the airmass of each target.

What is new in photometry?

- **Aperture and PSF photometry**: an empirical PSF for crowded fields such as globular clusters, and quality flags for every star.
- **Photometric calibration**: zero points, extinction and color terms from standard stars, with the Landolt and Clem & Landolt catalogues built in.
- **Color-magnitude diagrams**, in calibrated magnitudes and colors.
- **Surface photometry**: brightness profiles of clusters and galaxies, with King and Sérsic fits.
- **Exoplanet transits**: differential photometry, light curves and the fit of a transit model.
- **Color images**: combines any number of bands with any colors, with many options to make the image beautiful. It can also remove the stars, leaving only the nebula or galaxy behind them.
- **WCS preserved**: trimming, alignment and stacking now keep the astrometry of the images.



The Crab Nebula from raw VLT/FORS1 images in B, V and R, with the stars removed by easyspec.

Four new tutorials

Each tutorial starts from raw data of large telescopes (ESO/WFI, NTT/EFOSC2 and VLT/FORS1), included in the repository:

- **color images of a nebula**: the Crab Nebula, from the raw frames to the image above;
- **aperture photometry of an open cluster**: the Jewel Box (NGC 4755), from the reduction to the calibrated color-magnitude diagrams;
- **PSF photometry of a globular cluster**: M79 (NGC 1904), with its color-magnitude diagram and King profile;
- **an exoplanet transit**: differential photometry and the fit of the transit.

The tutorials and their data are available **on GitHub** at `https://github.com/ranieremenezes/easyspec/tree/main`, and the documentation of all functions at `https://easyspec.readthedocs.io`.